

UNZA: Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 11 (2020/2021)

1. The grouped frequency distribution below shows the masses, to the nearest gram, of 84 letters delivered by the postman.

Mass(g)	1 – 20	21 – 40	41 – 60	61 – 80	81 – 100
Number of letters	10	18	24	14	18

- (a) Complete the frequency distribution table below.

Mass(g)	Interval width	Frequency	Frequency density	Cumulative Frequency
$0.5 \leq x < 20.5$	20	10	0.5	10

- (b) Draw a frequency histogram to illustrate these data.

- (c) Which interval represents the modal class.

2. In one month, a student recorded the lengths, to the nearest minute, of each of the lectures she attended. The table below shows her data and the calculations she made before drawing a histogram to illustrate these data.

Length of lecture (minutes)	50 – 53	54 – 55	56 – 59	60 – 67
Number of lectures	a	b	30	c
Frequency density	5	13	7.5	1.5

Calculate

- (a) The value of a , of b and of c .

- (b) The number of lectures attended in a month.

3. (a) Two tutorials groups that wrote the same MAT1110 quiz 3 had the following results:

Group 9 : 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68

Group 10 : 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84

For each group, calculate

- i. Range.
- ii. Mean.
- iii. Mean deviation about the mean.

- (b) Calculate the mean for the following frequency distribution.

Mass(g)	1 – 20	21 – 40	41 – 60	61 – 80	81 – 100
Number of letters	10	18	24	14	18

- (c) Taking $A = 25.5$, as your assumed mean, calculate the mean for the following frequency distribution.

Classes	Frequency
1 – 10	11
10 – 20	29
20 – 30	18
30 – 40	4
40 – 50	5
50 – 60	3

4. (a)

Class A : 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68

Class B : 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84

- State the mode for each class.
- Calculate the median for each class.

(b) A group of University students took part in a sponsored race. The number of laps completed is given in the table below. Use the information to:

- Calculate an estimate for the mean number of laps.
- Determine the modal class.
- Determine the class interval containing the median.
- Compute the mode and the median.

Number of laps	Frequency (x)
1 – 5	2
6 – 10	9
11 – 15	15
16 – 20	20
21 – 25	17
26 – 30	25
31 – 35	2
36 – 40	1

5. (a) Two tutorials groups that wrote the same MAT1110 quiz 3 had the following results:

Group 5 : 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68

Group 6 : 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84

Calculate the variance and standard deviation for each group.

(b) Calculate the variance and standard deviation for the frequency distribution below.

Classes	Frequency
1 – 10	11
10 – 20	29
20 – 30	18
30 – 40	4
40 – 50	5
50 – 60	3